# PROGRESS LOG

PROGRESS LOG: WEEK 7-8

**1. What I've done:**

- Built the baseline Random Forest model, and performed SHAP analysis for all three risk classes, checked the results for representative Low, Medium and High risk points on waterfall plots.
- Completed training model, as supervisor suggested to extend the rainfall data from 3 to 40 years (1987-2025) and to retrain the model (99.62% accuracy); the elevation and distance to Clyde continued to be the top two predictors.
- Both versions of the dashboard (Version A and Version B) for the usability study were completed and correct with deployment on Local host; both contained the postcode search, compass indicator and sourced historical flood context, except that Version B included an explanation panel.
- Structured analysis codes into Data collection, EDA, Feature engineering, ML Model, Dashboard with numbered notebooks and started literature review on non-expert understanding of explainable AI.

**2. Issues or barriers:**

- The original 3-year rainfall window was found to be drier than the 39-year average, and used a non-representative daily maximum, which were identified and corrected.

Next two weeks planning:

**Technical:**

- Capture flood safety measures from sources on both versions of the dashboard; confirm past flood events with future risk areas.

**Ethics and Recruitment:**

- Already approved ethics; recruiting 6 participants, running usability sessions this week.

**Write-up:**

- Ongoing report review and writing of further chapters and methodology.
- The technical pipeline and the two versions of the dashboard are completed, with ethics approval to follow, the project is progressing well and is in sync with the scheduled 17 August submission.